Chapter 1

$$\begin{cases} 1C = 1A \times 1sec \\ 1e = 1.602\times10^{-19}C \\ 1C = 6.24\times10^{18}e \end{cases}$$

$$i = \frac{dq}{dt}$$

$$q(t) = \int_{t_0}^{t} i\,dt + q(0)$$

$$V = \frac{dw}{dq}$$

$$p = \frac{dw}{dt} = vi$$

$$w = \int_{t_0}^{t} p\,dt = \int_{t_0}^{t} vi\,dt$$

Chapter 2

§2.2 Ohm's Law $v = iR$

$G = \frac{1}{R}$ (S)

$$\begin{cases} P = vi = i^2R = \frac{v^2}{R} \\ P = vi = \frac{i^2}{G} = v^2G \end{cases}$$

§2.3

network - a circuit

branch, node, loop

mesh - loop that does not enclose any other loops.

$b = m + n - 1$

§2.4 Kirchhoff's Laws

KCL $\Sigma i = 0$

KVL $\Sigma v = 0$.

§2.5

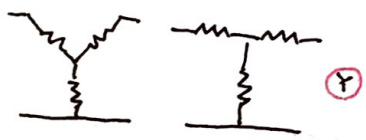
series: $R = \Sigma R_i$

principle of voltage divider: $v_i = \frac{R_i}{R} v$

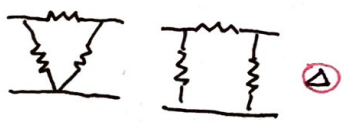
parallel: $G = \Sigma G_i$ ($\frac{1}{R} = \Sigma \frac{1}{R_i}$)

principle of current divider: $i_i = \frac{G_i}{G} i$

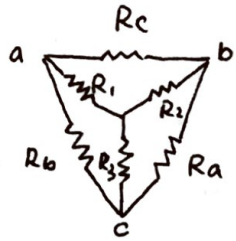
§2.7 Wye-Delta.



Y



Δ



Δ→Y

$$\begin{cases} R_1 = \frac{R_bR_c}{R_a+R_b+R_c} \\ R_2 = \frac{R_aR_c}{R_a+R_b+R_c} \\ R_3 = \frac{R_bR_c}{R_a+R_b+R_c} \end{cases}$$

Y→Δ

$$\begin{cases} R_a = \frac{R_1R_2+R_2R_3+R_1R_3}{R_1} \\ R_b = \frac{R_1R_2+R_2R_3+R_1R_3}{R_2} \\ R_c = \frac{R_1R_2+R_2R_3+R_1R_3}{R_3} \end{cases}$$

证明 对Y. 用KVL. R.
对Δ 用KCL. G
化出 v.

Chapter 3

§3.2 Nodal Analysis

① Select reference

② Assign $v_1, v_2, \dots, v_{n-1}$

③ KCL ④ Solve.

Nodal Analysis by Inspection

$$\begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1N} \\ G_{21} & G_{22} & \cdots & G_{2N} \\ \vdots & \vdots & & \vdots \\ G_{N1} & G_{N2} & \cdots & G_{NN} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix}$$

G_{kk} - sum of conductances connected to node k.

$G_{kj} = G_{jk}$ - negative of the sum of the conductances directly connecting nodes k & j ($k \neq j$)

i_k - sum of current sources entering node k.

*Inspection is only used in cases without corresponding sources.

§3.3 Nodal Analysis with Voltage Source

① reference ⊖ nonreference

② nonreference ⊖ noreference

⇒ Supernode (with elements in parallel)

§3.4 Mesh Analysis

① Assign mesh currents $i_1, i_2, \dots, i_n$.

② KVL ③ Solve.

Mesh Analysis by Inspection

$$\begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1N} \\ R_{21} & R_{22} & \cdots & R_{2N} \\ \vdots & \vdots & & \vdots \\ R_{N1} & R_{N2} & \cdots & R_{NN} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_N \end{bmatrix}$$

R_{kk} - sum of resistances in mesh k.

$R_{kj} = R_{jk}$ - negative of the sum of the resistances in common with meshes k & j ($k \neq j$).

V_k - sum taken clockwise of voltage sources in mesh k. with voltage rise treated as positive

§3.5 Mesh Analysis with Current Source

① ⊖ excusive

② ⊖ shared.

⇒ Supermesh (exclude elements in series with ⊖)

Nodal	Mesh
more parallel connected element	more series connected elements

mesh analysis

* planar

# of meshes = # of independent loops.

* 注意用大小写区分 current 和 mesh current

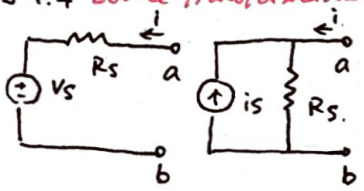
VE 215 Mid 1

Chapter 4

§4.2.4.3 Linearity & Superposition

* replace voltage sources with short cuts
* replace current sources with open circuits
* leave dependent sources.

§4.4 Source Transformation



$$\boxed{V_s = i_s R \text{ or } i_s = \frac{V_s}{R}}$$

§4.5 Thevenin's Theorem

V_{Th} - open-circuit voltage

R_{Th} - equivalent resistance when independent sources are turned off.

① No independent sources

$R_{Th} = R_{eq}$

② Independent sources.

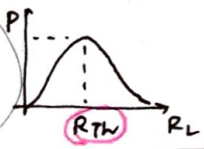
$R_{Th} = \frac{V_o}{i_o}$

§4.6 Norton's Theorem.

$R_N = R_{Th}$. $I_N = \frac{V_{Th}}{R_{Th}}$

§4.8 Maximum Power Transfer

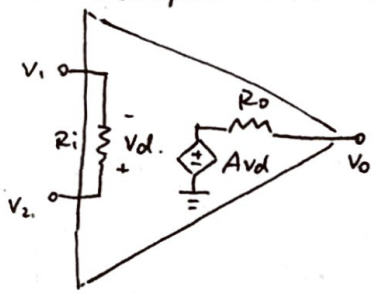
$P = i^2 R_L = \left(\frac{V_{Th}}{R_{Th}+R_L}\right)^2 R_L$



Chapter 5

§5.3 Ideal Operational Amplifier

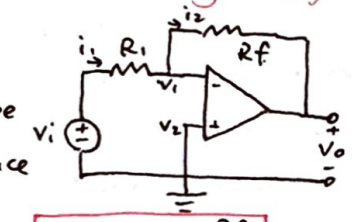
$A = \infty$ (open-loop gain)
$R_i = \infty$ (input resistance)
$R_o = 0$ (output resistance)



⇒ ① $i_1 = i_2 = 0$

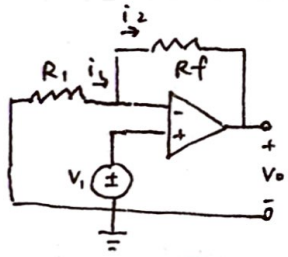
② $V_d = 0$

§5.4 Inverting Amplifier



$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{R_f}{R_1}}$$

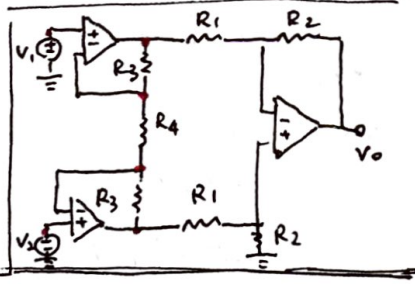
§5.5 Non-inverting Amplifier



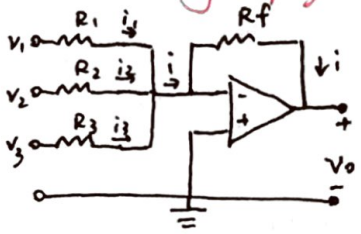
$$\boxed{A_v = \frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}}$$

$R_f = 0$ or $R_1 = \infty$. then $A_v = 1$.

voltage follower

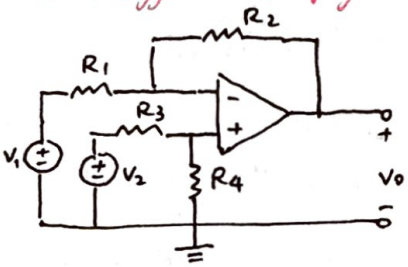


§5.6 Summing Amplifier



$$\boxed{V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)}$$

§5.7 Difference Amplifier

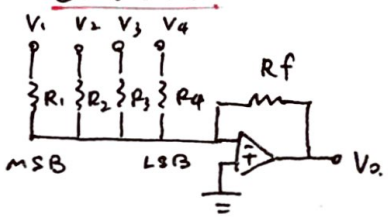


$$\boxed{V_o = \frac{R_1+R_2}{R_1}\,\frac{R_4}{R_3+R_4}V_2 - \frac{R_2}{R_1}V_1}$$

§5.8. Cascaded Op Amp Circuits

Stage.

§5.10.1 Digital-to-Analog Converter.



$$-V_o = \frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3 + \frac{R_f}{R_4}V_4$$

$$= k(2^3V_1 + 2^2V_2 + 2^1V_3 + 2^0V_4)$$

§5.10.2. Instrumentation Amplifiers.

$V_o = \frac{R_2}{R_1}\left(1 + \frac{2R_3}{R_4}\right)(V_2 - V_1)$

$R_4 = RG$. → can be used to adjust A.

$R_3 = R_2 = R_1 = R$, then

$V_o = A_v(V_2 - V_1) = A_v V_d$.

$V_d = V_2 - V_1$

$V_c = \frac{V_1 + V_2}{2}$.

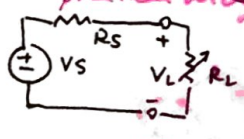
⇔ $V_2 = \frac{1}{2}V_d + V_c$

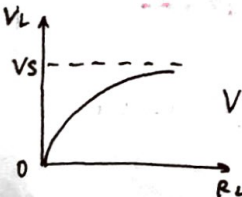
$V_1 = -\frac{1}{2}V_d + V_c$.

* input resistance is high

Source modeling (§4.10)

practical voltage source





$V_L = \frac{R_L}{R_s + R_L}V_s$.

① $R_s \ll R_L$ ideal

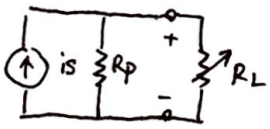
② open-circuited $V_{oc} = V_s$.

unloaded

with R_L. $V_L \downarrow$.

loading effect

practical current source



$i_L = \frac{R_p}{R_p + R_L}i_s$

① $R_p \gg R_L$ ideal

② short-circuited $i_{sc} = i_s$

unloaded

with R_L, $i_L \downarrow$

loading effect